Implementing a 1D PRD source function

Full wavelength- and height-dependent $S(\lambda, z)$ applied to 3D opacity

Implementation plan · 2026-09-08

Idea

For a scattering resonance line, *opacity is local* (keep computing it from the real 3D T, v, populations) but the *source function is non-local*; borrow the full PRD $S(\lambda, z)$ from the 1D NLTE model and integrate the *real* formal solution:

$$I_\lambda = \sum_i \left(1 - e^{-\Delta\tau_{\lambda,i}}\right) e^{-\tau_{\lambda,<i}}\, S(\lambda', h_i'), \qquad \eta_i(\lambda) = \chi_i(\lambda)\, S(\lambda', h_i').$$

A **single** source function is used — *no line/continuum split.* Off limb the continuum contributes a negligible pedestal, and near line core $\chi_{\rm line} \gg \chi_{\rm cont}$, so the total $S = \eta/\chi$ from the file is already the line value there; applying it to the total 3D opacity is accurate across the window and removes the continuum-isolation step entirely.

The curved ray already yields the geometric height h_i of every sample point, and the tangent height sets the minimum, so S is sampled over $h \geq h_{\rm tangent}$ — precisely the off-limb regime. The lookup coordinates are velocity- and calibration-shifted: $\lambda' = \lambda - \Delta\lambda_D(v_i)$, $h_i' = h_i - \Delta z_{\rm cal}$ (§4–5).

Build the PRD table — once, on rank 0

- Load `disk_center_test_opem.fits` $\Rightarrow$ $\chi(z, \lambda)$, $\eta(z, \lambda)$, $z_{\rm 1D}$ (82 pts), $\lambda_{\rm 1D}$ (1001). Interpolate in λ onto the synthesis grid `wave` so the two grids are guaranteed aligned.
- Form the total source function directly: $S(\lambda, z) = \eta(\lambda, z)/\chi(\lambda, z)$ — no subtraction, no guard (χ never vanishes thanks to the continuum floor).
- Result: a compact table {`lam_1d, z_1d, S[Nλ,Nz]`} (∼0.66 MB).

Distribute to workers via MPI broadcast (adjustment 1)

Only rank 0 reads the file. Before the manager/worker loop, all ranks call once

```
s1d = comm.bcast(s1d, root=0)
```

(rank 0 passes the table, others pass `None`); each worker holds it for the whole run. This costs one transfer per worker (∼0.66 MB, negligible) and **no worker opens the file on disk**. It is independent of the existing tag-based `send`/`recv` task machinery.

Apply $S(\lambda, z)$ in the LOS integration (adjustment 2)

- `create_ray`: also carry the height — add `'Height': z` (km) to `param_ray` (already computed, currently discarded).
- `calc_op_em`: compute the total opacity $\chi(\lambda, z)$ as now (line from pops + LTE continuum). Build the lookup coordinates per sample point, $\lambda_j' = \lambda_j - \Delta\lambda_D(v_i)$ (same $\Delta\lambda_D = (v_i/c)\lambda_0$ already used for the profile φ) and $h_i' = h_i - \Delta z_{\rm cal}$; evaluate `RegularGridInterpolator((lam_1d, z_1d), S)` at all (λ_j', h_i') $\Rightarrow$ `S_ray[Nλ,Ns]`. Then set $\eta = \chi \cdot$ `S_ray` (the population-based emissivity is no longer used for S). Return χ, η.
- `synth`: call `simple_formal_solution(op, em, ds)` and **return the real** I instead of $1 - e^{-\tau}$; keep the old behaviour behind a flag for side-by-side comparison.
- `worker_work(rank, s1d)` / `synth(..., s1d)`: thread the broadcast table through. MPI overseer/worker structure is otherwise unchanged.

Height calibration — free parameter (adjustment 3)

$\Delta z_{\rm cal}$ is a single scalar aligning MURaM's $z = 0$ (τ_{500}=1) with FALC's $z = 0$, with convention $z_{\rm 1D}^{\rm lookup} = h_{\rm ray} - \Delta z_{\rm cal}$. Exposed as a top-level CLI argument (`sys.argv`), default 0, and applied *at lookup time in the worker* — so the broadcast table stays offset-agnostic and one run can sweep $\Delta z_{\rm cal}$ without rebuilding it. Calibrate by matching the disk-centre profile or the limb-intensity inflection.

Caveats

- FALC tops out at **2.34 Mm**; set `bounds_error=False` and fill above the top with S at the top (or 0). Tall spicules reach higher — an extended/semi-empirical 1D model may be needed for the high LOS.
- The velocity shift assumes S co-moves with the plasma (PRD in the atom frame); it can be disabled for a rest-frame first run.
- Line/continuum split is deliberately neglected here (justified off limb); revisit only if a wing or on-disk comparison later needs it.
- The broken `take_given_S` branch (undefined `otherids`) is replaced by this path.

Scope: `create_ray`, `calc_op_em`, `synth`, `worker_work`, and `__main__` (load + `bcast` + new $\Delta z_{\rm cal}$ CLI arg).